

08.2 — Test Cases

For: Tester / admin running any test cycle on Haritha Hospitals ERPNext. **Goal:** Catalog of runnable test cases — copy, adapt, execute. **Last updated:** 2026-08-29
Pre-req reading: 08.1 Test Plan, 08.3 Regression Test Script

Test ID conventions

- **T-SMOKE-NNN** — smoke test
 - **T-EMP-NNN** — Employee DocType
 - **T-SHIFT-NNN** — Shift Type / Shift Assignment
 - **T-ATT-NNN** — Attendance
 - **T-LEAVE-NNN** — Leave Application / Allocation
 - **T-PAY-NNN** — Payroll / Salary Slip
 - **T-ROSTER-NNN** — Roster SPA
 - **T-INT-NNN** — Integration (cross-module)
 - **T-CUST-NNN** — Customization (CF/PS/PF/LH)
 - **T-MIG-NNN** — Migration / data import
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Smoke tests

Run first, ~5 min. All must pass before proceeding to deeper tests.

T-SMOKE-001: Login as Administrator

- **Priority:** High
- **Preconditions:** pberpprod reachable, Administrator user exists
- **Steps:**
 1. Navigate to <https://pberpprod.duckdns.org>
 2. Enter administrator / admin password
 3. Click Login
- **Expected:** Home dashboard loads in <5 sec; no error toast
- **Pass criteria:** Sidebar visible, “Welcome Administrator” message

T-SMOKE-002: View Employee list (200+ records load)

- **Priority:** High
- **Preconditions:** Logged in as Administrator or HR Manager
- **Steps:**
 1. Human Resources → Employee → List
 2. Wait for list to load
- **Expected:** List view shows 200+ records; pagination works; filter dropdown responsive
- **Pass criteria:** No 500 error, load time <10 sec

T-SMOKE-003: View Shift Type list (with colors)

- **Priority:** High
- **Preconditions:** haritha_hospital installed
- **Steps:**
 1. Human Resources → Shift Type → List

- 2. Open any shift (e.g. G0900R0900)
- **Expected:** List shows 25+ shifts; color property visible (hex code in field)
- **Pass criteria:** Shift opens in <3 sec; color field shows hex like #1f77b4

T-SMOKE-004: View Roster for current month

- **Priority:** High
- **Preconditions:** Logged in; employees have shift assignments
- **Steps:**
 1. Navigate to <https://pberpprod.duckdns.org/hr/roster>
 2. Default to current month
- **Expected:** Roster calendar renders with shift colors (blue/green/orange/violet); no JS console errors
- **Pass criteria:** Roster SPA loads, color legend visible, filter works

T-SMOKE-005: Submit attendance

- **Priority:** High
- **Preconditions:** Logged in as employee with active shift
- **Steps:**
 1. Human Resources → Attendance → New
 2. Fill employee, date = today, status = Present
 3. Save → Submit
- **Expected:** DocStatus = 1 (Submitted); appears in list
- **Pass criteria:** No validation errors; doc in submitted state

T-SMOKE-006: Apply for leave

- **Priority:** High
- **Preconditions:** Employee has Casual Leave balance > 1
- **Steps:**
 1. Human Resources → Leave Application → New
 2. Fill leave type = CL, from date = tomorrow, to date = tomorrow
 3. Submit
- **Expected:** Leave Application in Open state; assigned to manager in ToDo
- **Pass criteria:** Approval workflow triggered

T-SMOKE-007: Generate payslip

- **Priority:** High
 - **Preconditions:** Employee has Salary Structure assigned, current month open
 - **Steps:**
 1. Human Resources → Salary Slip → New (or via Payroll Entry)
 2. Fill employee, month = current
 3. Save
 - **Expected:** Salary Slip with earnings/deductions/net pay
 - **Pass criteria:** PF, ESI, PT, TDS computed; net pay > 0
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Functional tests — Employee module

T-EMP-001: Create new Employee with all required fields

- **Priority:** High
- **Preconditions:** Administrator logged in, Department (Cardiology) and Designation (Staff Nurse) exist
- **Steps:**
 1. Navigate to HR > Employee > New
 2. Fill First Name, Last Name, Company (Haritha Hospitals), Department (Cardiology), Designation (Staff Nurse)
 3. Set Date of Joining = today
 4. Save
 5. Verify Employee record created
- **Expected:** Employee appears in list; custom fields visible (Blood Group, Emergency Contact, etc.)
- **Pass criteria:** DocStatus = 0; reload shows all fields

T-EMP-002: Edit employee's cell number (allowed field)

- **Priority:** High
- **Preconditions:** Employee exists
- **Steps:**
 1. Open Employee record
 2. Edit cell_number field
 3. Save
- **Expected:** Save succeeds; audit log entry
- **Pass criteria:** No permission error

T-EMP-003: Edit employee's date of joining (forbidden field)

- **Priority:** High
- **Preconditions:** Employee exists; non-admin user
- **Steps:**
 1. Log in as HR User (not Manager)
 2. Open Employee record
 3. Attempt to edit date_of_joining
- **Expected:** Field is read-only (Property Setter); save fails or field disabled
- **Pass criteria:** Property Setter read_only enforced

T-EMP-004: Mark employee inactive

- **Priority:** Medium
- **Preconditions:** Employee active
- **Steps:**
 1. Open Employee record
 2. Set status = Inactive, relieving_date = today
 3. Save
- **Expected:** Status changes; excluded from active employee lists
- **Pass criteria:** List filter status = Active no longer shows this employee

T-EMP-005: Bulk create 10 employees via fixture

- **Priority:** Medium

- **Preconditions:** Test fixture with 10 employees JSON
- **Steps:**
 1. bench --site X execute haritha_hospital.tests.fixtures.bulk_create_employees
 2. Verify count via `frappe.db.count("Employee")`
- **Expected:** 10 new employees created; no duplicates if re-run
- **Pass criteria:** Idempotent (re-run produces 0 new)

T-EMP-006: View employee reporting hierarchy

- **Priority:** Medium
- **Preconditions:** Multiple employees with reports_to set
- **Steps:**
 1. Open Employee A (reports to B)
 2. Click B's name → opens B's record
 3. B's record shows "X direct reports"
- **Expected:** Hierarchy navigable; count accurate
- **Pass criteria:** Counts match `frappe.db.sql("SELECT COUNT(*) FROM tabEmployee WHERE reports_to='B'")`

T-EMP-007: Search employee by name

- **Priority:** Low
- **Preconditions:** 200+ employees
- **Steps:**
 1. Top search bar → type partial name (e.g. "Venka")
 2. Wait for suggestions
- **Expected:** Matching employees appear in dropdown
- **Pass criteria:** Search returns <2 sec; first match opens correct record

T-EMP-008: Print employee details (Letter Head)

- **Priority:** Low
- **Preconditions:** Haritha Hospitals Letter Head configured
- **Steps:**
 1. Open Employee → Print → PDF
- **Expected:** PDF shows Letter Head (logo + address) on every page
- **Pass criteria:** Letter Head visible, no overlap with content

T-EMP-009: Custom field 'blood_group' on Employee renders

- **Priority:** High
- **Preconditions:** haritha_hospital installed; Employee.blood_group CF exists
- **Steps:**
 1. Open any Employee record
 2. Look for blood_group field in Personal section
- **Expected:** Field visible with Select dropdown (A+, A-, B+, B-, AB+, AB-, O+, O-)
- **Pass criteria:** Field renders; saves correctly

T-EMP-010: Custom field 'emergency_contact_number' saves

- **Priority:** High
- **Steps:**
 1. Open Employee → edit emergency_contact_number

- 2. Save
- **Expected:** Saved; appears in audit log diff
- **Pass criteria:** Reload preserves value

T-EMP-011: Employee date of joining validation

- **Priority:** Medium
- **Preconditions:** HR Manager
- **Steps:**
 1. Create Employee with date_of_joining in the future
 2. Save
- **Expected:** Validation error "Date of Joining cannot be in future"
- **Pass criteria:** Save blocked, error message displayed

T-EMP-012: Employee gender field required

- **Priority:** Medium
- **Steps:**
 1. Create Employee, leave gender empty
 2. Save
- **Expected:** Validation error "Gender is mandatory"
- **Pass criteria:** Save blocked

T-EMP-013: Relieving date before date of joining

- **Priority:** Medium
- **Steps:**
 1. Open Employee with DOJ = 2020-01-01
 2. Set relieving_date = 2019-01-01
 3. Save
- **Expected:** Validation error "Relieving Date before Date of Joining"
- **Pass criteria:** Save blocked

T-EMP-014: Employee with same email as another user

- **Priority:** Medium
- **Steps:**
 1. Try to create Employee with user_id = existing user's email
- **Expected:** Validation error or uniqueness check
- **Pass criteria:** No duplicate user created

T-EMP-015: Delete employee with salary history

- **Priority:** Medium
- **Preconditions:** Employee has past Salary Slips
- **Steps:**
 1. Try to delete Employee
- **Expected:** Blocked (link integrity); or allowed with cascade warning
- **Pass criteria:** System prevents accidental data loss

T-EMP-016: Export employee list to CSV

- **Priority:** Low

- **Steps:**
 1. Employee list → Menu → Export
 2. Choose fields, download
- **Expected:** CSV file with selected columns
- **Pass criteria:** All rows present; headers match selection

T-EMP-017: Employee birthday today indicator

- **Priority:** Low
- **Preconditions:** At least one employee with DOB = today
- **Steps:**
 1. Home dashboard → check birthday widget
- **Expected:** Today's birthdays listed
- **Pass criteria:** Employee name appears; clickable to profile

T-EMP-018: Filter employee by department + designation

- **Priority:** Low
- **Steps:**
 1. Employee list → add filter: Department = Cardiology, Designation = Senior Consultant
- **Expected:** Filtered list shows only matching
- **Pass criteria:** Count matches manual SQL

T-EMP-019: Employee onboarding checklist completion

- **Priority:** Medium
- **Steps:**
 1. Open newly created Employee
 2. Verify all required fields populated
- **Expected:** No "Missing mandatory field" warnings
- **Pass criteria:** Print/PDF view clean

T-EMP-020: Deactivate employee preserves history

- **Priority:** High
 - **Steps:**
 1. Deactivate Employee X (status = Inactive)
 2. Run past Salary Slip report
 3. Verify X still appears in past slips
 - **Expected:** Past data preserved; only future transactions excluded
 - **Pass criteria:** Historical reports unaffected
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Functional tests — Shift module

T-SHIFT-001: Create new Shift Type

- **Priority:** High
- **Preconditions:** Administrator
- **Steps:**
 1. Human Resources → Shift Type → New
 2. Fill name = TEST9999R0800, start_time = 09:00:00, end_time = 17:00:00, duration = 8

- 3. Save
- **Expected:** Shift created; is_past_end_time auto-set if end < start
- **Pass criteria:** Reload shows shift

T-SHIFT-002: Shift Type naming convention check

- **Priority:** High
- **Steps:**
 1. Try to create Shift Type with name Invalid Name With Spaces
 2. Save
- **Expected:** Either accepted (Frappe allows) or blocked (custom validator)
- **Pass criteria:** Behavior consistent with masters/shift_type.csv convention

T-SHIFT-003: Shift Type last_sync_of_checkin required

- **Priority:** High (LEARNINGS #42)
- **Steps:**
 1. Create Shift Type without last_sync_of_checkin
 2. Enable mark_auto_attendance
 3. Save
- **Expected:** Warning or save proceeds; auto-attendance may not work without this field
- **Pass criteria:** Document behavior in fixture

T-SHIFT-004: Shift Assignment bulk create

- **Priority:** High
- **Steps:**
 1. Shift Assignment → New
 2. Add 5 employees in child table, dates next week
 3. Submit
- **Expected:** 5 shift assignments created; visible on roster
- **Pass criteria:** Roster SPA shows the shifts in correct cells

T-SHIFT-005: Shift Assignment overlap validation

- **Priority:** Medium
- **Steps:**
 1. Create Shift Assignment for Employee A, dates 2026-09-01 to 2026-09-30
 2. Try to create another Shift Assignment for A, dates 2026-09-15 to 2026-10-15
- **Expected:** Validation error or auto-cancel of overlapping range
- **Pass criteria:** No double-assignment

T-SHIFT-006: Shift Type color renders on roster

- **Priority:** High
- **Steps:**
 1. Open roster for date with shift G0900R0900
 2. Verify cell color matches color property (blue #1f77b4)
- **Expected:** Blue cell visible
- **Pass criteria:** Color hex matches Property Setter

T-SHIFT-007: Shift Type delete with assignments

- **Priority:** Medium
- **Steps:**
 1. Try to delete Shift Type that has 5+ Shift Assignments
- **Expected:** Blocked with link integrity error
- **Pass criteria:** Existing assignments preserved

T-SHIFT-008: Shift Type list shows duration

- **Priority:** Low
- **Steps:**
 1. Shift Type list view
 2. Verify duration column visible
- **Expected:** All shifts show duration (e.g. 8.5 hours)
- **Pass criteria:** Computed value matches manual calc (end - start, with past-midnight handling)

T-SHIFT-009: Night shift crosses midnight

- **Priority:** Medium
- **Steps:**
 1. Create Shift Type N2200R0800 (22:00 to 06:00 next day, 8h duration)
 2. Verify is_past_end_time = 1 and duration = 8
- **Expected:** Both auto-set; roster shows on start date
- **Pass criteria:** Past-midnight logic correct

T-SHIFT-010: Holiday List assignment on Shift Type

- **Priority:** Medium
- **Steps:**
 1. Open Shift Type → set Holiday List = "Haritha Hospitals Holiday List"
 2. Save
- **Expected:** Holidays from list impact attendance on that shift
- **Pass criteria:** Attendance on holiday = Holiday (not Present)

T-SHIFT-011: Shift Type for inactive employee

- **Priority:** Medium
- **Steps:**
 1. Create Shift Assignment for an Inactive employee
- **Expected:** Allowed or blocked (per policy)
- **Pass criteria:** Behavior documented

T-SHIFT-012: Reassign shift mid-period

- **Priority:** Medium
- **Steps:**
 1. Open existing Shift Assignment (active)
 2. Change dates
 3. Save
- **Expected:** New dates saved; roster updates
- **Pass criteria:** Roster reflects change within 1 min

T-SHIFT-013: Cancel Shift Assignment

- **Priority:** Medium
- **Steps:**
 1. Open Shift Assignment → Cancel
 2. Submit
- **Expected:** DocStatus = 2 (Cancelled); roster removes the shift
- **Pass criteria:** Roster no longer shows

T-SHIFT-014: Shift Type enable_auto_attendance toggle

- **Priority:** Medium
- **Steps:**
 1. Open Shift Type → toggle enable_auto_attendance
 2. Save
 3. Create Employee Checkin for that shift
- **Expected:** When ON, auto-attendance marks Present; when OFF, manual only
- **Pass criteria:** Behavior matches toggle

T-SHIFT-015: Shift grace periods

- **Priority:** Medium
- **Steps:**
 1. Set begin_check_in_before_shift_start_time = 60 min
 2. Employee checks in 30 min before shift start
- **Expected:** Late entry flag NOT triggered
- **Pass criteria:** Grace respected

T-SHIFT-016: Shift Assignment with employee on leave

- **Priority:** Medium
- **Steps:**
 1. Employee X has approved leave on 2026-09-15
 2. Try to assign Shift for 2026-09-15 to X
- **Expected:** Allowed but roster shows conflict warning; or blocked
- **Pass criteria:** Visual indicator or validation

T-SHIFT-017: Bulk shift assignment via fixture

- **Priority:** Medium
- **Steps:**
 1. Run fixture loader: `bench --site X execute haritha_hospital.tests.fixtures.bulk_shift_as`
 2. Verify count = N
- **Expected:** N assignments created; idempotent
- **Pass criteria:** Re-run produces 0 new

T-SHIFT-018: Shift Type list sort by start_time

- **Priority:** Low
- **Steps:**
 1. Shift Type list → sort by start_time ascending
- **Expected:** Earliest shifts first (M0600R0830 before M0800R0830)
- **Pass criteria:** Sort works on all 25+ shifts

T-SHIFT-019: Shift Type is_oncall flag

- **Priority:** Low
- **Steps:**
 1. Open Shift Type → check is_oncall checkbox
 2. Save
- **Expected:** Flag saved; visible in custom reports
- **Pass criteria:** CF works

T-SHIFT-020: Shift Type is_emergency flag

- **Priority:** Low
 - **Steps:**
 1. Open Shift Type → check is_emergency
 2. Save
 - **Expected:** Flag saved; appears in emergency shift report
 - **Pass criteria:** CF works
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Functional tests — Attendance module

T-ATT-001: Submit attendance manually

- **Priority:** High
- **Steps:**
 1. HR → Attendance → New
 2. Fill employee, date = today, status = Present
 3. Save → Submit
- **Expected:** Attendance submitted, working_hours auto-filled if shift exists
- **Pass criteria:** DocStatus = 1

T-ATT-002: Duplicate attendance for same employee/date

- **Priority:** High
- **Steps:**
 1. Submit attendance for Employee A on 2026-09-15
 2. Try to submit another for A on same date
- **Expected:** Validation error “Attendance already exists”
- **Pass criteria:** Save blocked

T-ATT-003: Auto attendance from checkin

- **Priority:** High
- **Steps:**
 1. Create Employee Checkin for Employee A at 09:00 (within shift window)
 2. Wait for scheduler to run (or trigger manually)
 3. Verify Attendance auto-created
- **Expected:** Attendance record with status = Present, working hours computed
- **Pass criteria:** Auto-attendance job runs successfully

T-ATT-004: Attendance with no checkin

- **Priority:** High

- **Steps:**
 1. Employee has shift but no checkin recorded
 2. Run auto-attendance
- **Expected:** Attendance NOT created (depends on process_attendance_after config)
- **Pass criteria:** No phantom Present entries

T-ATT-005: Late entry flag

- **Priority:** Medium
- **Steps:**
 1. Shift start = 09:00, late_entry_grace = 15 min
 2. Employee checks in at 09:20
- **Expected:** Attendance shows late_entry = 1
- **Pass criteria:** Flag set

T-ATT-006: Half-day threshold

- **Priority:** Medium
- **Steps:**
 1. Shift = 8h, half_day threshold = 5h
 2. Employee works 4h (checkin 09:00, checkout 13:00)
- **Expected:** Attendance status = Half Day
- **Pass criteria:** Status correct

T-ATT-007: Absent threshold

- **Priority:** Medium
- **Steps:**
 1. Shift = 8h, absent threshold = 0h
 2. Employee checks in 09:00, no checkout
- **Expected:** Attendance status = Absent (working_hours = 0)
- **Pass criteria:** Status correct

T-ATT-008: Attendance on holiday

- **Priority:** Medium
- **Steps:**
 1. Holiday = 2026-09-15 (per Holiday List)
 2. Employee works that day, checkin at 09:00
- **Expected:** Attendance status = Holiday (if mark_auto_attendance_on_holidays = 0) or Present with comp-off (if =1)
- **Pass criteria:** Behavior matches config

T-ATT-009: Attendance on leave

- **Priority:** Medium
- **Steps:**
 1. Employee has approved leave on 2026-09-15
 2. Run auto-attendance for that date
- **Expected:** Attendance status = On Leave (not Present)
- **Pass criteria:** Leave overrides attendance

T-ATT-010: Manual attendance overrides auto

- **Priority:** Medium
- **Steps:**
 1. Auto-attendance creates Present record
 2. HR manually marks Absent for same date
 3. Verify manual wins
- **Expected:** Manual record is the source of truth
- **Pass criteria:** No duplicate; manual preserved

T-ATT-011: Attendance Report — present count

- **Priority:** Medium
- **Steps:**
 1. Run Attendance Summary report for Sept 2026
 2. Verify present count = sum of Present status rows
- **Expected:** Count matches SQL: `SELECT COUNT(*) FROM tabAttendance WHERE status='Present' AND MONTH(attendance_date)=9`
- **Pass criteria:** Report accurate

T-ATT-012: Attendance cancellation

- **Priority:** Medium
- **Steps:**
 1. Submit Attendance
 2. Cancel
- **Expected:** DocStatus = 2; cancelled entry visible in audit
- **Pass criteria:** Cancel succeeds

T-ATT-013: Attendance amend after cancel

- **Priority:** Low
- **Steps:**
 1. Cancel Attendance
 2. Amend (create new with same date)
- **Expected:** New attendance created with amended data
- **Pass criteria:** Amend workflow works

T-ATT-014: Attendance bulk import

- **Priority:** Medium
- **Steps:**
 1. Upload CSV with 100 attendance rows
 2. Verify all created via Data Import
- **Expected:** 100 records; no duplicates if re-run
- **Pass criteria:** Idempotent

T-ATT-015: Attendance working hours calc

- **Priority:** Medium
- **Steps:**
 1. Checkin 09:00, checkout 18:00 (9h shift)
 2. Verify working_hours = 9.0

- **Expected:** Calc matches First Check-in and Last Check-out
- **Pass criteria:** Field correct

T-ATT-016: Attendance for past date

- **Priority:** Medium
- **Steps:**
 1. Create Attendance for date 30 days ago
 2. Verify save (vs block if too old)
- **Expected:** Per policy — manager can mark 7 days back, HR 30 days
- **Pass criteria:** Permission enforced

T-ATT-017: Attendance deletion

- **Priority:** Medium
- **Steps:**
 1. Try to delete submitted Attendance
- **Expected:** Blocked (only Cancel allowed for submitted docs)
- **Pass criteria:** Integrity preserved

T-ATT-018: Attendance Request approval flow

- **Priority:** High
- **Steps:**
 1. Employee creates Attendance Request (correction)
 2. Manager approves
 3. Verify Attendance record auto-updated
- **Expected:** Request → Approval → Attendance update
- **Pass criteria:** End-to-end works

T-ATT-019: Attendance Request rejection

- **Priority:** Medium
- **Steps:**
 1. Manager rejects Attendance Request with reason
 2. Verify Attendance NOT modified
- **Expected:** Request status = Rejected; reason logged
- **Pass criteria:** No silent modification

T-ATT-020: Auto-attendance after long absence

- **Priority:** Low
 - **Preconditions:** Employee returned after 30-day absence
 - **Steps:**
 1. Set process_attendance_after = 2026-08-01
 2. Run auto-attendance for September
 - **Expected:** August attendance NOT processed (before cutoff)
 - **Pass criteria:** Backfill logic correct
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Functional tests — Leave module

T-LEAVE-001: Apply for Casual Leave

- **Priority:** High
- **Steps:**
 1. Leave Application → New
 2. Leave Type = CL, From = tomorrow, To = tomorrow, Reason = personal
 3. Submit
- **Expected:** Status = Open, balance reduced by 1 (allocated to pending)
- **Pass criteria:** Workflow triggered

T-LEAVE-002: Approve leave application

- **Priority:** High
- **Steps:**
 1. Manager opens pending Leave Application
 2. Approve
- **Expected:** Status = Approved, balance deducted (not just allocated)
- **Pass criteria:** Balance updates correctly

T-LEAVE-003: Reject leave application

- **Priority:** High
- **Steps:**
 1. Manager rejects with reason “insufficient coverage”
- **Expected:** Status = Rejected, balance restored
- **Pass criteria:** Reason captured, balance correct

T-LEAVE-004: Leave with insufficient balance

- **Priority:** High
- **Steps:**
 1. CL balance = 0
 2. Apply for 1 day CL
- **Expected:** Validation error “Insufficient balance”
- **Pass criteria:** Save blocked

T-LEAVE-005: Half-day leave

- **Priority:** Medium
- **Steps:**
 1. Apply for half-day CL on 2026-09-15
 2. Submit
- **Expected:** 0.5 day deducted from balance
- **Pass criteria:** Decimal handling correct

T-LEAVE-006: Overlapping leave applications

- **Priority:** Medium
- **Steps:**
 1. Apply for CL 2026-09-15 to 2026-09-17
 2. Submit (approved)
 3. Try to apply for CL 2026-09-16 to 2026-09-18

- **Expected:** Validation error “Overlaps with existing leave”
- **Pass criteria:** Save blocked

T-LEAVE-007: Leave type with max consecutive days

- **Priority:** Medium
- **Steps:**
 1. CL max = 3 days at a time
 2. Apply for 5 days
- **Expected:** Validation error “Max 3 consecutive”
- **Pass criteria:** Save blocked

T-LEAVE-008: Sick leave requires medical certificate

- **Priority:** Medium
- **Steps:**
 1. SL > 2 days → must attach medical certificate
 2. Apply for 4 days SL without attachment
- **Expected:** Validation error or warning
- **Pass criteria:** Policy enforced

T-LEAVE-009: Earned Leave accrual

- **Priority:** Medium
- **Steps:**
 1. Verify monthly accrual: 1.5 days per month
 2. Run Leave Allocation job
- **Expected:** EL balance increases by 1.5 each month
- **Pass criteria:** Ledger entry created

T-LEAVE-010: Leave carry forward calculation

- **Priority:** Medium
- **Steps:**
 1. EL balance = 40 on 2026-03-31
 2. Run year-end process
- **Expected:** Balance = 30 (max carry forward) on 2026-04-01; 10 days encashed or lapsed per policy
- **Pass criteria:** Logic correct

T-LEAVE-011: Comp-Off credit for holiday work

- **Priority:** Medium
- **Steps:**
 1. Employee works on holiday 2026-09-15 (Sunday)
 2. HR credits Comp-Off
 3. Verify Comp-Off balance +1
- **Expected:** Comp-Off allocated; visible in leave balance
- **Pass criteria:** Ledger entry created

T-LEAVE-012: Leave calendar view shows team

- **Priority:** Low

- **Steps:**
 1. Leave Application → Calendar view
 2. Filter by Department
- **Expected:** Calendar shows approved leaves for department
- **Pass criteria:** Visual rendering correct

T-LEAVE-013: Leave cancellation by employee

- **Priority:** Medium
- **Steps:**
 1. Open approved Leave Application
 2. Cancel (allowed before leave starts)
- **Expected:** Status = Cancelled; balance restored
- **Pass criteria:** Cancel allowed pre-start

T-LEAVE-014: Leave ledger entry audit

- **Priority:** Medium
- **Steps:**
 1. Open any employee's Leave Type → Ledger tab
 2. Verify all transactions logged with timestamp, amount, type
- **Expected:** Every allocation / submission / cancellation creates a ledger entry
- **Pass criteria:** Audit trail complete

T-LEAVE-015: Maternity leave 26 weeks

- **Priority:** Medium
 - **Preconditions:** Female employee
 - **Steps:**
 1. Apply for Maternity Leave 182 days (26 weeks)
 2. Submit
 - **Expected:** Allowed (policy)
 - **Pass criteria:** Saved without error
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Functional tests — Payroll module

T-PAY-001: Salary Slip basic generation

- **Priority:** High
- **Steps:**
 1. Salary Slip → New
 2. Employee with Salary Structure, month = current
 3. Save
- **Expected:** Slip generated with earnings + deductions + net pay
- **Pass criteria:** All components computed

T-PAY-002: PF deduction = 12% of Basic

- **Priority:** High
- **Steps:**
 1. Employee Basic = ₹50,000
 2. Generate Salary Slip

- **Expected:** PF = ₹6,000
- **Pass criteria:** Calculation accurate

T-PAY-003: ESI applies for gross < ₹21,000

- **Priority:** High
- **Steps:**
 1. Employee gross = ₹18,000
 2. Generate slip
- **Expected:** ESI = $0.75\% \times 18000 = ₹135$
- **Pass criteria:** ESI computed

T-PAY-004: ESI does NOT apply for gross $\geq$ ₹21,000

- **Priority:** High
- **Steps:**
 1. Employee gross = ₹25,000
 2. Generate slip
- **Expected:** ESI = 0
- **Pass criteria:** Threshold correct

T-PAY-005: Professional Tax for Karnataka

- **Priority:** Medium
- **Steps:**
 1. Generate slip for Karnataka employee
- **Expected:** PT = ₹200 (state default)
- **Pass criteria:** State-specific PT applied

T-PAY-006: TDS monthly computation

- **Priority:** High
- **Steps:**
 1. Employee annual income > tax slab
 2. Generate monthly slip
- **Expected:** TDS = annual tax / 12 (approximate)
- **Pass criteria:** Monthly TDS deducted

T-PAY-007: Loss of pay for unpaid leave

- **Priority:** High
- **Steps:**
 1. Employee on 3 days LWP in month
 2. Generate slip
- **Expected:** LWP deduction = $3 \times (\text{gross} / \text{working_days})$
- **Pass criteria:** LWP correctly computed

T-PAY-008: Process Payroll batch

- **Priority:** High
- **Steps:**
 1. Payroll Entry → New
 2. Fill company, month, all employees

- 3. Create Salary Slips
- **Expected:** Batch of salary slips created; submission status updates
- **Pass criteria:** All employees processed

T-PAY-009: Salary Slip cancellation

- **Priority:** Medium
- **Steps:**
 1. Submit Salary Slip
 2. Cancel (HR Manager only)
- **Expected:** Status = Cancelled; reversal entry in ledger
- **Pass criteria:** Cancel succeeds with permission

T-PAY-010: Print Salary Slip PDF

- **Priority:** Medium
- **Steps:**
 1. Open Salary Slip → Print → PDF
- **Expected:** PDF with Haritha letter head, salary table, net pay highlighted
- **Pass criteria:** PDF renders correctly

T-PAY-011: Salary Structure assignment

- **Priority:** Medium
- **Steps:**
 1. Create Salary Structure “Nurse L1”
 2. Assign to 5 employees
- **Expected:** All 5 show assigned structure in profile
- **Pass criteria:** Assignment persists

T-PAY-012: Bank advice generation

- **Priority:** Low
- **Steps:**
 1. Payroll Entry → Bank Advice
- **Expected:** CSV with employee, account, net pay, IFSC
- **Pass criteria:** Bank-ready file generated

T-PAY-013: Payroll anomaly report

- **Priority:** Medium
- **Steps:**
 1. Run “Payroll Anomaly” report for last month
- **Expected:** Lists employees with variance > threshold
- **Pass criteria:** Report identifies outliers

T-PAY-014: Form 16 generation

- **Priority:** Medium
- **Steps:**
 1. Payroll → Form 16 → select employee + FY
- **Expected:** Form 16 PDF generated per Indian tax format
- **Pass criteria:** Format compliant

T-PAY-015: Full & Final Settlement

- **Priority:** Medium
 - **Preconditions:** Relieved employee
 - **Steps:**
 1. Run F&F process for resigned employee
 - **Expected:** Final slip includes: pro-rated salary + leave encashment - notice period recovery - loans
 - **Pass criteria:** Net pay correct per formula
-

Functional tests — Roster SPA

T-ROSTER-001: Roster loads for current month

- **Priority:** High
- **Steps:**
 1. Navigate to /hr/roster
 2. Default to current month
- **Expected:** Calendar renders with shift colors
- **Pass criteria:** No JS errors, <5 sec load

T-ROSTER-002: Filter by Department

- **Priority:** High
- **Steps:**
 1. Roster → Department filter = Cardiology
- **Expected:** Only Cardiology employees shown
- **Pass criteria:** Filter applied correctly

T-ROSTER-003: Filter by single employee

- **Priority:** Medium
- **Steps:**
 1. Roster → search → select Employee X
- **Expected:** Only X's shifts shown
- **Pass criteria:** Single-employee view works

T-ROSTER-004: Date picker navigation

- **Priority:** Medium
- **Steps:**
 1. Click "next month" arrow
- **Expected:** Roster updates to next month
- **Pass criteria:** Navigation works for ±12 months

T-ROSTER-005: View toggle (Day / Week / Month)

- **Priority:** Low
- **Steps:**
 1. Toggle between Day, Week, Month views
- **Expected:** Each view renders correctly with shifts
- **Pass criteria:** All 3 views functional

T-ROSTER-006: Color legend visible

- **Priority:** Low
- **Steps:**
 1. Open roster → verify legend (G=blue, M=green, A=orange, N=violet)
- **Expected:** Legend with 4 colors and labels
- **Pass criteria:** Legend matches shift colors

T-ROSTER-007: Hover shows shift details

- **Priority:** Low
- **Steps:**
 1. Hover any shift cell
- **Expected:** Tooltip with shift name, start, end, employee name
- **Pass criteria:** Tooltip within 500ms

T-ROSTER-008: Click cell opens employee profile

- **Priority:** Low
- **Steps:**
 1. Click any shift cell
- **Expected:** Modal or navigation to Employee record
- **Pass criteria:** Click navigates correctly

T-ROSTER-009: Roster export to PDF

- **Priority:** Low
- **Steps:**
 1. Roster → Menu → Export PDF
- **Expected:** PDF with calendar grid + shift colors
- **Pass criteria:** PDF renders

T-ROSTER-010: Roster performance with 250 employees

- **Priority:** Medium
 - **Steps:**
 1. Open roster with full Haritha workforce (250)
 2. Verify load time <5 sec
 - **Expected:** SPA responsive
 - **Pass criteria:** No lag, no errors
-

Integration tests

T-INT-001: Employee Checkin → Auto Attendance

- **Priority:** Critical
- **Steps:**
 1. Employee X has Shift Assignment for today
 2. Insert Employee Checkin record at 09:00 (within shift)
 3. Run auto-attendance job
 4. Verify Attendance auto-created with status = Present
- **Expected:** End-to-end: Checkin → Attendance

- **Pass criteria:** Attendance record exists with working_hours computed

T-INT-002: Attendance → Payroll

- **Priority:** Critical
- **Steps:**
 1. Employee X has full month attendance
 2. Run Process Payroll
 3. Verify Salary Slip includes correct working_days, no LWP
- **Expected:** Attendance feeds into payroll
- **Pass criteria:** Slip working_days = actual present count

T-INT-003: Shift Assignment → Roster view

- **Priority:** Critical
- **Steps:**
 1. Create Shift Assignment for Employee X for next week
 2. Open Roster SPA filtered by X
 3. Verify shift appears in correct day cell with color
- **Expected:** SPA reflects DB
- **Pass criteria:** Real-time sync (or within cache TTL)

T-INT-004: Leave Approval → Attendance deduction

- **Priority:** Critical
 - **Steps:**
 1. Employee X approves leave for 2026-09-15
 2. Run auto-attendance for 2026-09-15
 3. Verify Attendance = On Leave (not Present)
 - **Expected:** Leave overrides attendance generation
 - **Pass criteria:** Status = On Leave
-

Customization tests

T-CUST-001: Custom Field on Employee visible

- **Priority:** High
- **Steps:**
 1. Open Employee record
 2. Verify blood_group, emergency_contact_number, etc. visible
- **Expected:** All CFs rendered
- **Pass criteria:** Reload after bench migrate preserves

T-CUST-002: Property Setter (Shift Type color) renders

- **Priority:** High
- **Steps:**
 1. Open Shift Type list
 2. Verify color field shows hex
 3. Open roster, verify cells match
- **Expected:** PS drives roster SPA colors
- **Pass criteria:** Color sync works

T-CUST-003: Print Format renders correctly

- **Priority:** Medium
- **Steps:**
 1. Open Salary Slip → Print
 2. Verify Print Format applied (custom HTML)
- **Expected:** Custom PF (not stock) renders
- **Pass criteria:** PF in fixtures matches expected

T-CUST-004: Letter Head displays on PDF

- **Priority:** Medium
 - **Steps:**
 1. Print any document with Letter Head
 2. Verify logo + address on PDF
 - **Expected:** LH visible on every page
 - **Pass criteria:** PDF includes LH
-

Migration tests

T-MIG-001: Idempotent re-run produces no duplicates

- **Priority:** High
- **Steps:**
 1. Run migration script (e.g. load 100 employees from CSV)
 2. Count: should be 100
 3. Re-run same script
 4. Count: still 100 (no duplicates)
- **Expected:** Idempotent
- **Pass criteria:** INSERT IGNORE or upsert logic works

T-MIG-002: Link integrity preserved

- **Priority:** High
- **Steps:**
 1. Before migration: count Shift Assignments linked to existing Shifts
 2. Run migration
 3. After: count should be equal (no orphans)
- **Expected:** Link fields updated correctly
- **Pass criteria:** No LinkValidationError

T-MIG-003: Counter docstatus preserved

- **Priority:** High
 - **Steps:**
 1. Before migration: count submitted vs draft Leave Applications
 2. Run migration
 3. After: counts match
 - **Expected:** DocStatus preserved
 - **Pass criteria:** Submission state intact
-

Where to next

- **Test plan?** See 08.1 Test Plan
- **Run regression?** See 08.3 Regression Test Script
- **Found a bug?** Open GitHub Issue with severity, steps, evidence